

## Exercise 1 — Employee Database (JDBC + MySQL)

**Goal:** Use Java JDBC to perform basic operations on a MySQL database.

### Requirements

1. **Create the database and table in MySQL:**
  - Create a database named EmployeeDB.
  - Create a table Employees with the following columns:
    - id (int, auto-increment, primary key)
    - name (varchar(50))
    - age (int)
    - department (varchar(50))
    - salary (double)
2. **Write a Java program that implements the following functions:**
  - Connect to the EmployeeDB database.
  - Add a new employee to the Employees table.
  - Display the list of all employees.
  - Update an employee's information by id.
  - Delete an employee by id.
  - Search for an employee by name or department.

## Exercise 2 — Product Database (JDBC + MySQL)

### Requirements

**Create the database and table in MySQL:** 1. Create a database named ProductDB. 2. Create a table Products with the following columns: - id (int, auto-increment, primary key) - name (varchar(100)) - category (varchar(50)) - price (double) - quantity (int)

**Write a Java program that implements the following functions:** 1. Connect to the ProductDB database. 2. Add a new product to the Products table. 3. Display the list of all products. 4. Update a product's information by id. 5. Delete a product by id. 6. Search for a product by name or category.

**Requirements:** - The in-memory data structure used to hold products before printing must use a Hashtable, with the key being the product's id and the value being the product object. - **Advanced (bonus):** implement the program using the DAO pattern (model, repository, and implementation layers).